

TrayGuard: What We Learned About Why Tray Errors Persist

EC ENGR 180DW — Final Design Review

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The Problem

The Problem By The Numbers

9 per 100 cases

Alfred et al. (2021)
3,900 defects in 41,799 cases
55% at assembly

10 min avg delay

Nichol et al. (2024)
236 errors, 147 cases
\$6–9M/yr estimated cost

44% wrong spec

Zhu et al. (2019)
398 errors in 33,839 packages
largest single category

These are recurring, measured operational failures — not isolated accidents.

Why Tray Errors Persist

What Tray Reconstruction Actually Requires

FIGURE TBD

figures/workstation'placeholder.pdf

- Identify each instrument by sight (no labels)
- Match against a count sheet with **local names**
- Distinguish lookalikes (straight vs curved Mayo)
- Check for damage, wear, contamination
- Count correct quantities
- All under **production pressure**

Six Root Causes

FIGURE TBD

figures/fishbone`placeholder.pdf

Three structural drivers:

- ① SPD evolved from materials management → **no licensure**
- ② Cost-center accounting → errors are **invisible**
- ③ No information loop → errors **repeat**

The Broken Feedback Loop

FIGURE TBD

[figures/feedback'loop'placeholder.pdf](#)

“Error reporting is cumbersome, human-dependent, delayed, and incomplete.” — Nichol et al.

We Evaluated Seven Solution Categories

Solution	Buildable?	Validatable?	Infra?
CV tray checker	No	No	Maybe
Physical error-proofing	No	No	No
Workflow redesign	No	No	No
Decision support	Part	Part	Yes
Task simplification	No	No	No
Accountability system	No	No	No
Training	Yes	Yes	Yes

Training is a Buildable Entry Point — Not a Cure

What training addresses

- Measurable local tray knowledge
- Lookalike discrimination practice
- Pre/post learning evidence

What training does NOT fix

- Broken count sheets
- Staffing shortages
- Structural wage problems
- The broken feedback loop

Training is the one piece we can build and validate **now**.

Why Computer Vision Alone Won't Work

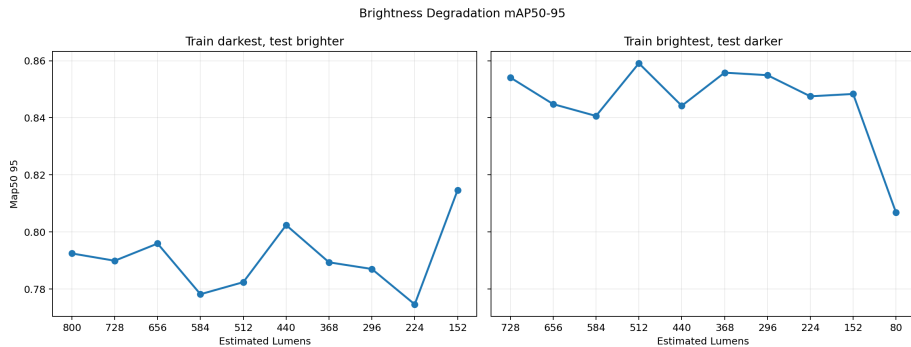
We Ran 7 Staged Detection Experiments

Experiment	Test	Key result
Bright → dim	Light tolerance	mAP50-95 = 0.781
Matte → reflective	Background	mAP50-95 = 0.759
Separated → overlay	Occlusion	mAP50-95 = 0.389
Real instruments	Shape vs position	Per-class AP: 0.000–0.862

Lavado baseline (clean data): **mAP50 = 0.990**

Pipeline works — the data doesn't generalize.

Brightness Degradation



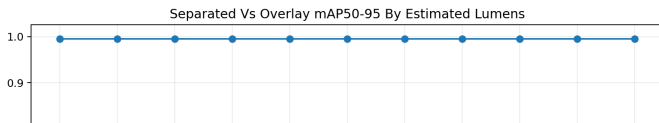
Occlusion Is the Dealbreaker



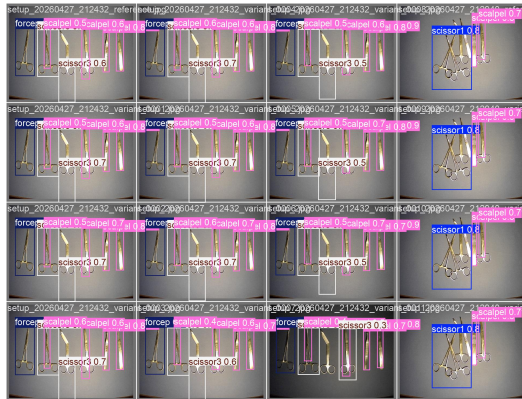
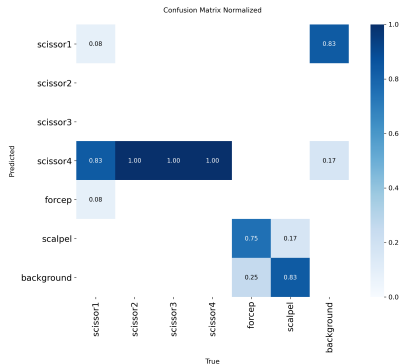
Separated layout (training)



Overlay layout (test)



Real Instruments: The Model Learned Position, Not Shape



Split	Surviving classes	Zero AP
Matte → reflective	scissor2, forcep, scalpel	scissor1/3/4
Reflective → matte	scalpel only	all others
Shape similarity	scissor1, scissor4	scissor2/3, forcep

The Cumulative Lesson

What a deployment-grade CV system still needs

- × Site-specific training data (instruments, lighting, layouts)
- × Cross-manufacturer validation [?]
- × Open-set rejection (unknown objects = false positives)
- × Human-in-the-loop for every uncertain case
- × Workflow integration in live SPD
- × Regulatory / device policy / privacy review

The training path doesn't need any of this for a first study.

Why Training Is the Best Entry Point

CV and Training Are Complementary, Not Competing

FIGURE TBD

[figures/complementary'placeholder.pdf](#)

The module is the durable artifact. The training loop proves the module works. The module then enables CV.

The Durable Artifact: Tray Module Schema

```
{
  "id": "mayo_scissors_straight_55",
  "display_name": "Straight Mayo, 5.5 in",
  "aliases": ["straight Mayo",
              "suture scissors"],
  "distinguishing_features": [
    "Heavy straight blades"
  ],
  "common_confusions": [
    "mayo_scissors_curved_55"
  ],
  "apriltag_id": 12,
  "study_prompts": [ ... ]
}
```

- Mirrors real count-sheet fields
- Adds aliases, lookalikes, prompts
- File-backed, versioned
- **Software is throwaway.**
- **Schema survives.**

Printed Cards: How It Works Physically

Front (assessment)

FIGURE TBD

figures/card`front`placeholder.pdf

Back (study)

FIGURE TBD

figures/card`back`placeholder.pdf

Neutral front prevents answer leakage. Tag ID maps to instrument ID.

The Scoring Architecture: 5 Error Categories

Category	Definition	Example
Correct	Right item, right count	1 scalpel handle
Missing	Required item not selected	No Kelly clamp
Wrong	Item not on tray list	Retractor not in this tray
Extra	Distractor selected	Item from different tray
Misidentified	Right class, wrong variant	Curved instead of straight
Wrong count	Right item, wrong quantity	2 Kellys, need 4

- Maps to literature error categories [? ? ?]
- Can separate *what* improved (fewer misses? fewer lookalikes?)

FIGURE TBD

figures/confidence/mockup/placeholder.pdf

High-confidence error

Learner was sure but wrong
Targeted feedback needed

Low-confidence correct

Learner was unsure but right
Fragile knowledge, needs reinforcement

The 7-Step Learning Loop

FIGURE TBD

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Pre-test (baseline) → Study cards → Quiz → Practice sort → Post-test → Export

The Research Question

How can a local tray training and assessment system help novice users improve their ability to detect missing, wrong, extra, misidentified, or miscounted instruments before supervised SPD work?

That question is narrow, testable, and honest about what we can claim.

What a Future Team Should Do

The Study Ladder

FIGURE TBD

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Kirkpatrick Level 2 target. Level 3–4 require future validation.

Study 1: Novice Pre/Post Design

Segment	Time	Activity
Consent + intake	3–5 min	Background
Pre-test tray sort	5–8 min	No hints, no feedback
Retrieval cards	10–12 min	Prompt-before-reveal
Quiz	5–8 min	With confidence
Practice sort	8–10 min	Error feedback
Post-test	5–8 min	Parallel variant, no hints
Photo-ID test	5–7 min	Convergent validity

n = 8–12 novices, one 60-min session, within-subject

Study 2: Delayed Retention

FIGURE TBD

figures/retention`placeholder.pdf

Criterion 1

$\geq 50\%$ of gain retained

Criterion 2

Accuracy $> 5\text{pp}$ above baseline

Criterion 3

$\geq 70\%$ return rate

If these fail, the claim reverts to “immediate learning only.”

Study 3: Expert Review

- Q Are instrument names and aliases correct for local practice?
- Q Are counts and distractors plausible for training?
- Q Are lookalike pairs educationally meaningful?
- Q Are feedback messages misleading or safe?
- Q Would the weak-item report help target coaching?

The main weakness of a student-built module is **content validity**. Expert review closes that gap.

Go/No-Go Decision

FIGURE TBD

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Study passes → SPD pilot is justified.
Study fails → problem may be harder than training alone.

Conclusion

What We Know and What We Don't

What we know

- Tray errors are systemic and well-documented
- CV alone has too high a validation burden
- Training is the buildable entry point
- The module schema is durable infrastructure
- The study ladder has falsification criteria

What we can't claim (yet)

- Training transfers to real SPD work
- System works for experienced techs
- It reduces OR delays
- CV is deployment-ready
- Same-day gains persist (Study 2)

Our project found that this problem is harder and more systemic than any single technical fix can address.

Our contribution is documenting *why*, and giving the next team a validated starting point — not a finished product.

The study is designed. The schema is defined. The scoring is tested.
Run the study.

